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Cambridge Assessment International Education

# Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*9288105350*

**CHEMISTRY** **5070/21**

Paper 2 Theory **May/June 2025**

**1 hour 45 minutes**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

**INFORMATION**

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.

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[Turn over

**1** Choose from the following salts to answer the questions.

**aluminium chloride**

**barium sulfate**

**calcium chloride**

**copper(II) sulfate**

**magnesium chloride**

**potassium iodide**

**potassium manganate(VII)**

**silver nitrate**

**sodium bromide**

**sodium sulfite**

Each salt can be used once, more than once or not at all.

State which salt:

**(a)** is prepared using a precipitation reaction

..... [1]

**(b)** in aqueous solution, reacts with an excess of aqueous ammonia to give a dark blue solution

..... [1]

**(c)** reacts with warm aqueous sodium hydroxide and aluminium foil to give a gas that turns damp red litmus paper blue

..... [1]

**(d)** has an aqueous solution that is used to test for an oxidising agent

..... [1]

**(e)** has an aqueous solution that reacts with copper metal.

..... [1]

[Total: 5]

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**2** A concentrated aqueous solution of copper(II) bromide is electrolysed using graphite electrodes.

**(a)** Graphite has good electrical conductivity.

**(i)** Explain why graphite has good electrical conductivity.

Use ideas about structure and bonding.

.....

.....

.....

..... [2]

**(ii)** State one **other** property of graphite that makes it suitable for use as an electrode during electrolysis.

..... [1]

**(b)** Predict the products of the electrolysis of concentrated aqueous copper(II) bromide with graphite electrodes.

product at anode .....

product at cathode .....
[2]

**(c)** Dilute sulfuric acid is electrolysed using graphite electrodes to form oxygen and hydrogen.

Construct the ionic half-equation for the reaction at each electrode.

reaction at anode .....

reaction at cathode .....
[2]

**(d)** Hydrogen and oxygen are used in a fuel cell to produce electricity.

**(i)** Name the only chemical product formed in a fuel cell.

..... [1]

**(ii)** Describe **one** disadvantage of using hydrogen–oxygen fuel cells in vehicles compared to gasoline or petrol engines.

.....

..... [1]

[Total: 9]

**3** The equation for the reaction between methane and chlorine is shown in Fig. 3.1.

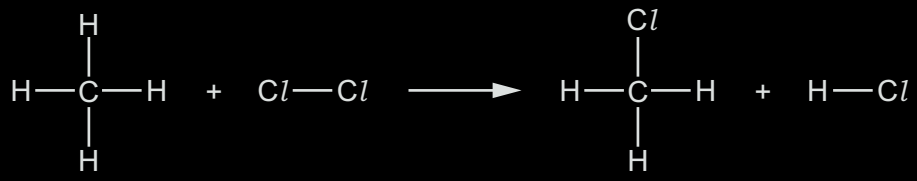


**Fig. 3.1**

**(a)** State **one** condition for this reaction.

..... [1]

**(b)** Explain why this reaction is an example of substitution.

.....

..... [1]

**(c)** Table 3.1 shows some bond energies.

**Table 3.1**

| bond | bond energy in kJ/mol |
|---|---|
| C—H | 410 |
| C—C*l* | 340 |
| C*l*—C*l* | 242 |
| H—C*l* | 431 |

Show by calculation that the enthalpy change of the reaction between methane and chlorine, $\Delta H$, is $-119\,kJ/mol$.

[3]

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**(d)** Complete the reaction pathway diagram in Fig. 3.2 for the reaction between methane and chlorine.

Label the:

- reactants
- products
- enthalpy change of the reaction, $\Delta H$
- activation energy, $E_a$.

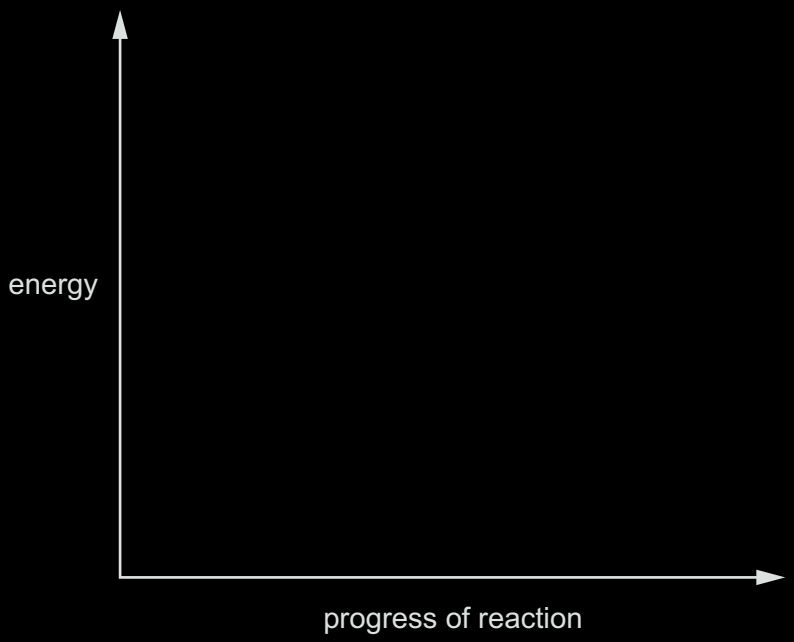



**Fig. 3.2**

[3]

**(e)** Draw a dot-and-cross diagram to show the electronic configuration in a molecule of methane.

Show only the outer shell electrons.

[1]

[Total: 9]

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**4** Ethanol, $C_2H_5OH$, is a member of the homologous series of alcohols.

**(a)** Give the general formula of the homologous series of alcohols.

..... [1]

**(b)** Members of a homologous series have the same general formula and share similar chemical properties.

State two **other** general characteristics of a homologous series.

1 .....

2 .....

[2]

**(c)** The equation for the reaction between ethanol and sodium is shown.

$$2Na + 2C_2H_5OH \rightarrow H_2 + 2C_2H_5ONa$$

A sample of 1.35 g of sodium is added to excess ethanol.

**(i)** Calculate the volume of hydrogen formed measured at room temperature and pressure.

Give your answer to **two** significant figures.

volume = ..... $dm^3$ [3]

**(ii)** Water is added to the reaction mixture to make an aqueous solution.

A few drops of litmus are then added. The litmus changes colour to blue.

Suggest the name of the ion present in the aqueous solution responsible for the colour change.

..... [1]

**(d)** State **two** uses for ethanol.

1 .....

2 .....

[2]

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**(e)** Describe the manufacture of ethanol from ethene.

Include the other reactant and the conditions for the manufacture.

.....

.....

.....

.....

..... [3]

[Total: 12]

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**5** Vehicles that use petrol as a fuel produce several air pollutants.

**(a)** Petrol is a mixture of hydrocarbons which includes octane, $C_8H_{18}$.

Explain why octane is a hydrocarbon.

.....

..... [1]

**(b)** Two of the air pollutants produced are carbon monoxide and nitrogen monoxide.

**(i)** Explain how carbon monoxide, CO, is formed in a petrol engine.

Include a symbol equation.

.....

.....

..... [2]

**(ii)** Explain how nitrogen monoxide, NO, is formed in a petrol engine.

Include a symbol equation.

.....

.....

..... [2]

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**(c)** A catalytic converter removes most of the CO and NO formed in a petrol engine.

$$2CO(g) + 2NO(g) \rightarrow 2CO_2(g) + N_2(g)$$

**(i)** Explain why this reaction involves both oxidation **and** reduction.

.....

.....

.....

..... [2]

**(ii)** The reaction is catalysed using platinum metal.

Explain how a catalyst increases the rate of a reaction.

.....

..... [1]

**(iii)** State and explain the effect of increasing the temperature on the rate of this reaction.

.....

.....

..... [2]

**(iv)** State and explain the effect of decreasing the pressure on the rate of this reaction.

.....

.....

..... [2]

[Total: 12]

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**6** Chlorine, $Cl_2$, is in Group VII of the Periodic Table.

**(a)** The melting point of chlorine is –101 °C and the boiling point is –35 °C.

**(i)** Explain why chlorine is a liquid at –50 °C.

.....

..... [1]

**(ii)** Describe the arrangement and motion of chlorine molecules at –50 °C.

.....

.....

.....

..... [3]

**(b)** A sample of chlorine gas contains $1.204 \times 10^{20}$ molecules.

One mole of chlorine gas contains $6.02 \times 10^{23}$ molecules.

Calculate the mass of this sample of chlorine gas.

mass of chlorine = ..... g [2]

**(c)** The ionic equation for the reaction of chlorine with cold dilute aqueous sodium hydroxide is shown.

$$Cl_2 + 2OH^- \rightarrow ClO^- + Cl^- + H_2O$$

**(i)** State the oxidation number of chlorine in $Cl_2$ and in $Cl^-$.

$Cl_2$ .....

$Cl^-$ .....

[2]

**(ii)** During the reaction chlorine is reduced.

Explain why, using ideas about electrons.

.....

..... [1]

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**(d)** Chlorine reacts with cold water to form an equilibrium mixture containing the acids HC$l$(aq) and HOC$l$(aq).

The forward reaction releases thermal energy into the surroundings.

$$Cl_2(aq) + H_2O(l) \rightleftharpoons HCl(aq) + HOCl(aq)$$

**(i)** The temperature of the equilibrium mixture is increased.

State and explain what happens to the acidity of the equilibrium mixture.

statement .....

explanation .....

.....

.....

[2]

**(ii)** HC$l$(aq) is a strong acid and HOC$l$(aq) is a weak acid.

Describe the difference between a strong acid and a weak acid.

.....

.....

..... [2]

**(e)** A chloride of iron contains 34.5% iron by mass.

Calculate the empirical formula of this chloride.

Show your working.

empirical formula ..... [3]

[Total: 16]

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**7** Aluminium is used in the manufacture of aircraft and food containers.

Aluminium is resistant to corrosion by water and oxygen.

**(a)** Give one **other** reason why aluminium is used in the manufacture of aircraft.

..... [1]

**(b)** Explain why aluminium is resistant to corrosion by water and oxygen.

.....

.....

..... [2]

**(c)** Aluminium metal reacts with hot dilute sulfuric acid to form hydrogen and aqueous aluminium sulfate as the only products.

Construct the symbol equation for this reaction.

Include state symbols.

..... [2]

**(d)** Aluminium oxide reacts with sulfuric acid and with the alkali aqueous sodium hydroxide.

State the name of the type of oxide that reacts with both acids and alkalis.

..... [1]

[Total: 6]

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**8** Fig. 8.1 shows the displayed formula of methylbut-2-enoate.

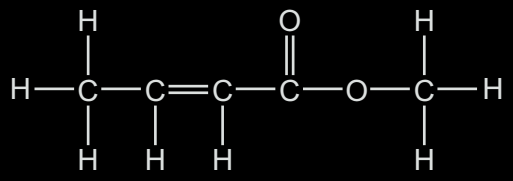


**Fig. 8.1**

**(a)** Methylbut-2-enoate is an unsaturated ester.

**(i)** Explain why methylbut-2-enoate is unsaturated.

.....

..... [1]

**(ii)** Describe a chemical test to show that methylbut-2-enoate is unsaturated.

.....

.....

..... [2]

**(b)** Methylbut-2-enoate is made by the reaction of an alcohol and a carboxylic acid in the presence of a catalyst.

**(i)** State the name of the type of catalyst used in this reaction.

..... [1]

**(ii)** State the name of the alcohol used in this reaction.

..... [1]

**(iii)** Draw the displayed formula of the carboxylic acid used in this reaction.

[1]

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**(c)** Methylbut-2-enoate is a monomer used to make an addition polymer.

**(i)** Draw the structure of this addition polymer.

Include at least **two** repeat units.

[2]

**(ii)** A sample of 80 g of methylbut-2-enoate is reacted to make the addition polymer.

There is a 100% yield.

State the mass of addition polymer made.

Explain your answer.

mass of addition polymer = ..... g

explanation .....

.....

[2]

**(d)** Fig. 8.2 shows compound **B**.

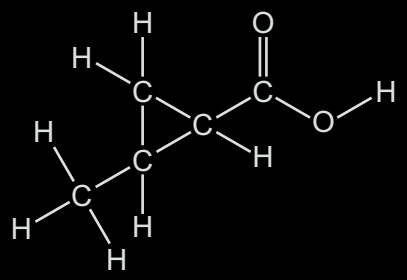



**Fig. 8.2**

Explain why methylbut-2-enoate and compound **B** are a pair of structural isomers.

.....

..... [1]

[Total: 11]

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# The Periodic Table of Elements

| Group I | Group II | | | | | | | | | | | Group III | Group IV | Group V | Group VI | Group VII | Group VIII |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | 1 H hydrogen 1 | | | | | | | | | | 2 He helium 4 |
| 3 Li lithium 7 | 4 Be beryllium 9 | | | | | | | | | | | 5 B boron 11 | 6 C carbon 12 | 7 N nitrogen 14 | 8 O oxygen 16 | 9 F fluorine 19 | 10 Ne neon 20 |
| 11 Na sodium 23 | 12 Mg magnesium 24 | | | | | | | | | | | 13 Al aluminium 27 | 14 Si silicon 28 | 15 P phosphorus 31 | 16 S sulfur 32 | 17 Cl chlorine 35.5 | 18 Ar argon 40 |
| 19 K potassium 39 | 20 Ca calcium 40 | 21 Sc scandium 45 | 22 Ti titanium 48 | 23 V vanadium 51 | 24 Cr chromium 52 | 25 Mn manganese 55 | 26 Fe iron 56 | 27 Co cobalt 59 | 28 Ni nickel 59 | 29 Cu copper 64 | 30 Zn zinc 65 | 31 Ga gallium 70 | 32 Ge germanium 73 | 33 As arsenic 75 | 34 Se selenium 79 | 35 Br bromine 80 | 36 Kr krypton 84 |
| 37 Rb rubidium 85 | 38 Sr strontium 88 | 39 Y yttrium 89 | 40 Zr zirconium 91 | 41 Nb niobium 93 | 42 Mo molybdenum 96 | 43 Tc technetium – | 44 Ru ruthenium 101 | 45 Rh rhodium 103 | 46 Pd palladium 106 | 47 Ag silver 108 | 48 Cd cadmium 112 | 49 In indium 115 | 50 Sn tin 119 | 51 Sb antimony 122 | 52 Te tellurium 128 | 53 I iodine 127 | 54 Xe xenon 131 |
| 55 Cs caesium 133 | 56 Ba barium 137 | 57–71 lanthanoids | 72 Hf hafnium 178 | 73 Ta tantalum 181 | 74 W tungsten 184 | 75 Re rhenium 186 | 76 Os osmium 190 | 77 Ir iridium 192 | 78 Pt platinum 195 | 79 Au gold 197 | 80 Hg mercury 201 | 81 Tl thallium 204 | 82 Pb lead 207 | 83 Bi bismuth 209 | 84 Po polonium – | 85 At astatine – | 86 Rn radon – |
| 87 Fr francium – | 88 Ra radium – | 89–103 actinoids | 104 Rf rutherfordium – | 105 Db dubnium – | 106 Sg seaborgium – | 107 Bh bohrium – | 108 Hs hassium – | 109 Mt meitnerium – | 110 Ds darmstadtium – | 111 Rg roentgenium – | 112 Cn copernicium – | 113 Nh nihonium – | 114 Fl flerovium – | 115 Mc moscovium – | 116 Lv livermorium – | 117 Ts tennessine – | 118 Og oganesson – |

**Key**

atomic number
atomic symbol
name
relative atomic mass

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| lanthanoids | 57 La lanthanum 139 | 58 Ce cerium 140 | 59 Pr praseodymium 141 | 60 Nd neodymium 144 | 61 Pm promethium – | 62 Sm samarium 150 | 63 Eu europium 152 | 64 Gd gadolinium 157 | 65 Tb terbium 159 | 66 Dy dysprosium 163 | 67 Ho holmium 165 | 68 Er erbium 167 | 69 Tm thulium 169 | 70 Yb ytterbium 173 | 71 Lu lutetium 175 |
| actinoids | 89 Ac actinium – | 90 Th thorium 232 | 91 Pa protactinium 231 | 92 U uranium 238 | 93 Np neptunium – | 94 Pu plutonium – | 95 Am americium – | 96 Cm curium – | 97 Bk berkelium – | 98 Cf californium – | 99 Es einsteinium – | 100 Fm fermium – | 101 Md mendelevium – | 102 No nobelium – | 103 Lr lawrencium – |

The volume of one mole of any gas is 24 $dm^3$ at room temperature and pressure (r.t.p.).



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